Introduction to Statistics and Biometry

Lecture Number 1

September 19, 2016

Outline of Presentation

1. Definition of Statistics
2. Definition of Biometry
3. Branches of Statistics
4. Definition of Basic Terms used in Statistics
5. Types of Variables and Data
6. Types of Data Sets
7. Parametric and Non-Parametric Statistics

Definition of Statistics

- Statistics is a very broad subject, with applications in a vast number of different fields.
- Originating as a science of statehood, statistics has found applications in areas such as Economics, Engineering, Business, Sociology, Psychology, Planning, Production, Education, Medicine, Agriculture, Forestry and Ecology.
- To date, there is no other human sphere where statistics cannot be applied.
- Statistics may be broadly defined as follows:

Statistics is a branch of mathematics which is concerned with scientific methods for collecting, organising, summarising, presenting and analysing data as well as deriving valid conclusions from the data and making reasonable decisions on the basis of this analysis.

Definition of Statistics...Cont'd

- Statistics may also be defined as a set of scientific principles and techniques that are useful in reaching conclusions about populations and processes when the available information is both limited and variable.
- In simple terms, statistics is the science of learning from data .
- We approach the study of statistics by considering the four steps in learning from data:
  1. Collection of data - it is the first step and this is the foundation upon which the entire data set lies. Careful planning is essential before collecting the data.
  2. Presentation of the data - the mass data collected should be presented in a suitable, concise form for further analysis.
  3. Analysis of the data - the data presented should be carefully analysed for making inferences such as measures of central tendencies, dispersion, correlation, regression etc.,
  4. Interpretation of the data - the final step is drawing conclusions from the data collected. A valid conclusion must be drawn on the basis of analysis.

Definition of Biometry

- The word Biometry may be viewed as comprising of two words, "bio-" which simply implies - connected with life and living things and "metric" which is associated with a system of measurement.
- Nevertheless, Biometry may be defined as follows:

Definition of Biometry

Biometry is the application of statistics in the biological sciences such as Agriculture, Forestry, Ecology, Wildlife Biology, Animal Science, Medicine and Pharmacy and so on.The subject is concerned with the design of research studies (experiments and surveys), data collection and management, data analysis and interpretation in order to address issues in biological systems.

Why should you study statistics?

- We hardly talk without making reference to numbers, rates or confidence.
- All the statements with reference to numbers, rates or confidence are meant to garner support for some assertions.
- One can conclude that we are living in a world of numbers.
- Thus, even a rudimentary knowledge of statistics will make you better able to evaluate statistical claims made by reporters, weather forecasters, radio and television advertisers, manufacturer's claims, political candidates, government officials, and other persons who may use statistics in the information or arguments they present.
- Statistical procedures are basic to research - which you will be expected to conduct before completing your studies. You may also be called on to conduct research or interpret research findings in your field after completion of studies.
- In everyday affairs, decisions are made based on the available data or information at hand.

Branches/Types of Statistics

- Statistics is sometimes divided into two main areas depending on how data are used. The two areas are:
  1. Descriptive Statistics
  2. Inferential Statistics

Descriptive statistics consists of methods for collection, organising, summarising and presentation of data. Descriptive statistics are typically presented graphically, in tabular form (in tables), or as summary statistics (single values).

Inferential statistics consists of generalizing from samples to populations, performing estimations and hypothesis tests, determining relationships among variables, and making predictions. Inferential statistics uses **probability**.

- We will now define some of the basic concepts or terms used in statistics - some of which we have mentioned in our definitions so far!

Definition of Basic Terms used in Statistics

- Population - is a collection of all subjects (human or otherwise) that are of interest or being studied. A population can be infinite or finite.
- Sample - any subset or group of subjects selected from a population of interest.
- Parameter - a numerical or descriptive measure of a population. It is a **fixed** value and is usually **unknown**.
- Statistic - a numerical or descriptive measure of a sample. A statistic **varies** from sample to sample. Unlike a parameter, a statistic is **known** or can be determined.
- Generally, a given population parameter corresponds to a sample statistic.
- In general, we use Greek or capital letters for population parameters and lower case Roman letters to denote sample statistics.

Definition of Basic Terms used in Statistics..Cont'd

- Estimate - a sample statistic estimates a given population parameter. It is generally not possible, or very difficult, to calculate the value of a parameter directly. Thus, a sample statistic is calculated to correspond to a given parameter.
- Variable (Variate) - is a characteristic or attribute that changes or varies over time and/or for different individuals or objects under consideration.
- For example, body temperature is a variable that changes over time within a single individual; it also varies from person to person.
- Observation - this is a record of each subject in the sample, e.g., measurement

Types of Variables and Data

- Data - are the values (measurements or observations) that the variables can assume.
- The variables whose values are determined by chance are called random variables.
- A collection of data values forms a data set. Each value in the data set is called a data value or a datum.
- Thus, to gain knowledge about seemingly haphazard situations, statisticians collect information for variables, which describe the situation.
- In nature, different sorts of variables are encountered and thus it's desirable to distinguish among them as illustrated in the Figure on the next slide.

Types of Variables and Data

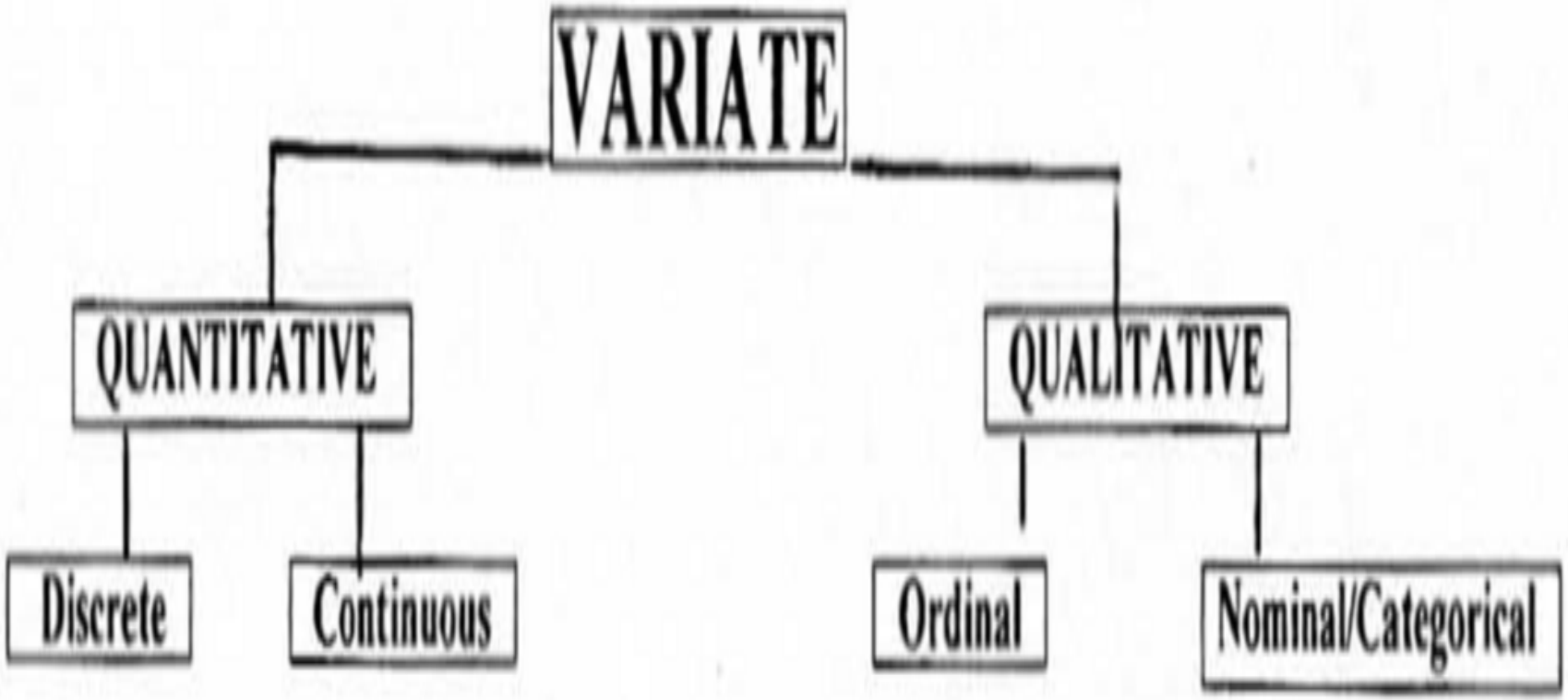


Figure: Types of Variables and Data

Types of Variables and Data

- Qualitative variables are variables that can be placed into distinct categories, according to some characteristic or attribute. That is, a qualitative variable describes a characteristic or attribute of an individual or object. It is a non-numeric entity.
- Thus, a qualitative variable has states or forms, which can be described, categorized, classified, or qualified. e.g. gender, species.
  1. Ordinal - deals with relative differences; e.g. high, medium, low
  2. Nominal/Categorical - associated with some quality or attribute which the variate posses. e.g. sex of an organism, name of species, colour of timber, type of vehicle.
- Quantitative - variable where data represents the amount or quantity of something and measurement is made over a range of values. Age, weight, height, volume, prices, costs, sales, sizes, etc., are all quantitative variables.

Types of Variables and Data

- A quantitative variable may be continuous or discontinuous i.e. discrete.
  1. Discrete - variables assume values that can be counted. Only certain values are possible (usually consecutive integers), e.g number of: trees in a plot, timber, insects in an insect trap, fish in an aquarium, animals in a quadrant etc.
  2. Continuous - variable can assume an infinite number of values between any two specific values. They are obtained by measuring. They often include fractions and decimals.
- Sometimes a new variable may be created as a function of existing variables and/or by applying mathematical functions. Such kind of variables are referred to as derived variables.
- Examples of some common derived variables include: Ratios, Proportions, Percentages and Rates.

Types of Data

- The type of statistical analysis that is appropriate for a particular variable depends on the data type (scale of measurement) used for the random variable.
- There are four data types (or scales of measurement), namely nominal, ordinal, interval and ratio.
  (1) Norminal-scaled data - the nominal level of measurement classifies data into mutually exclusive (non-overlapping), exhausting categories in which no order or ranking can be imposed on the data.
    - ★ Examples are the labels sex, programme of study, species name, marital status, religious affiliation, employment status, education-level .
    - ★ These labels are only used to identify an attribute of the variable.
    - ★ Under nominal measurements, 'numbers' are simply to identify, classify, categorize or distinguish.
    - ★ For instance, scores 0 for male and 1 for female are simply used to identify, distinguish, categorize or classify the subject by sex.

Types of Data

- The next level of measurement is called the ordinal level.
  (2) Ordinal-scaled data - ordinal level of measurement classifies data into categories that can be ranked; however, precise differences between the ranks do not exist.
    - ★ Ordinal data are generated from ranked responses (also generated from a counting process).
    - ★ Examples include: grade (A,B,C,D), judging(first, second, third place), rating scale(excellent, good, poor), ranking of players etc.
    - ★ The observation for each variable possesses characteristics of nominal data in that each response rating is a label for excellent, good or poor quality.
    - ★ In addition, the data can be ranked with respect to quality.
    - ★ An ordinal variate may also be represented by scores such as 1 = poor, 2 = fair, 3 = good, etc.
    - ★ Thus, ordinal scaled-data have the properties of nominal data but the order or rank of the data is meaningful or important and data can be numeric or nonnumeric

Types of Data

- The third level of measurement is called the interval level.
  (3) Interval-scaled data - interval level of measurement ranks data, and precise differences between units of measure do exist; however, there is no meaningful zero. These are really numeric.
    - ★ This level differs from the ordinal level in that precise differences do exist between units.
    - ★ Temperature is one example of interval measurement, since there is a meaningful difference of $1^{\circ}C$ between each unit, such as 35 and $36^{\circ}C$.
    - ★ IQ is another example of such a variable. There is a meaningful difference of 1 point between an IQ of 109 and an IQ of 110.
    - ★ One property is lacking in the interval scale -there is no true zero.
    - ★ For example, IQ tests do not measure people who have no intelligence. For temperature, 0° does not mean no heat at all.

Types of Data

- The final and highest level of measurement is called the ratio level.
  (4) Ratio-scaled data - ratio level of measurement possesses all the characteristics of interval measurement, and there exists a true zero. In addition, true ratios exist when the same variable is measured on two different members of the population.
    - ★ Examples of ratio scales are those used to measure height, weight, area, age, time, money etc.
    - ★ Ratio scales have differences between units (1 inch, 1 kg, etc.) and a true zero.
    - ★ In addition, the ratio scale contains a true ratio between values.
    - ★ For example, if one person can lift 20 kg and another can lift 10 kg, then the ratio between them is 2 to 1 - put another way, the first person can lift twice as much as the second person.
- *Note: there is not complete agreement among statisticians about the classification of data into one of the four categories - e.g.some researchers classify IQ data as ratio data rather than interval.Also, data can be altered so that they fit into a different category.*

Types of Data Sets

- Data sets can be: Univariate, Bivariate or Multivariate
  (1) If you only measure or observe one variable per subject, the resulting data set is univariate. For example, if you just measure height for each of 10 selected students in class, the resulting 10 data values constitute a univariate data set.
  (2) If you measure or observe two variables per subject, the resulting data set is bivariate. For example, you could measure height and weight for each of 10 selected students in class.
  (3) If you measure or observe three or more variables on the same subject, the resulting data set is multivariate. For example, you could measure height, weight and blood pressure for each of 10 selected students in class.

Why should you be concerned about types of variables and data?

- Knowledge about different kinds of variables and the data that they generate is important because the methods used to describe data sets depend on the type of data you have collected.
- For each set of data that you collect, the key will be to determine what type of data you have and how you can present them most clearly and understandably to your audience!
- There are many statistical methods or techniques for analysing data and the choice depends, to a larger extent, on the type of variable(s) and data at hand.

Parametric and Non-Parametric Statistics

- Common statistical methods for scientists/biologists fall into 2 classes - parametric and non-parametric methods.
  (1) **Parametric Statistics** - deals with statistical methods which assume that sample data comes from a population that follows a probability distribution based on a fixed set of parameters.
    - It is the oldest and most often used by statisticians and researchers.
    - It is characterized with making strict assumptions, which may not always hold.
    - Thus, it may not always be the most appropriate for analysing biological data.

  (2) **Non-Parametric Statistics** - covers statistical methods that do not rely on data belonging to any particular probability distribution.
    - It is most recent and not based on making stringent assumptions about the data.
- Thus, we have parametric and non-parametric statistics - much of our learning will be devoted to parametric statistics.